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Public Investment Multipliers Across High- and Low-Debt States in the European Union

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Abstract

This paper studies how the effectiveness of public investment depends on the prevailing level of government debt. We develop a flexible-price real business cycle model with productive public capital, government debt, and fiscal feedback rules, calibrated to aggregated European Union data. The model is solved separately for two fiscal regimes—a low-debt period (1995–2007) and a high-debt period (2008–2024)—while holding preference and technology parameters constant across regimes. Public investment shocks are analyzed using impulse response functions and a range of output multipliers, and the model’s empirical performance is evaluated using the Simulated Method of Moments. The results show that public investment generates substantially larger and more persistent output responses in the low-debt regime, driven by stronger crowd-in effects on private investment and labor supply. In contrast, higher public debt dampens the transmission of public investment through stronger tax adjustments required to stabilize debt. By comparison, total factor productivity shocks produce nearly identical dynamics across regimes, highlighting that the observed heterogeneity is specific to fiscal policy rather than general business-cycle mechanisms. Overall, the findings underscore the importance of fiscal space in shaping the macroeconomic returns to public investment and suggest that high public debt can significantly limit the effectiveness of investment-led fiscal stimulus.

Keywords

Public investment

Fiscal policy

Government debt

DSGE models

Impulse Response Functions

Fiscal multipliers

Simulated Method of Moments

Business cycle dynamics

Debt-Regime dependence

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1 Introduction

Public investment is widely regarded as a powerful instrument for fostering long-term economic growth through the accumulation of productive capital. However, its impact is not always consistent and is depends critically on the current level of public debt at the time of policy implementation. With rising debt levels in the European Union, it becomes increasingly unclear whether public investment policies that were effective in the past would generate comparable outcomes today. High-debt environments constrain governments' fiscal capacity and often require stronger fiscal adjustment, potentially dampening the growth effects of public investment. Low-debt states, on the other hand, often enjoy larger fiscal space, allowing effective translation of public investment into sustained economic gains.

While a large literature studies fiscal multipliers, much of it abstracts from explicit debt-state dependence. Building on this gap, this paper examines how public investment effectiveness differs across high- and low-debt regimes within a Real Business Cycle (RBC) Framework. It addresses the question "How do Debt Levels affect the Effectiveness of Public Investment through the Investment Multiplier?" by employing EU-specific parameter calibration, examining the impulse response functions, and simulating the method of moments for external validity. The results indicate that public investment shocks generate substantially larger and more persistent multipliers in low-debt regimes, whereas productivity shocks produce broadly similar responses across debt states.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data used in the analysis. Section 4 presents the model and its equilibrium conditions. Section 5 discusses the calibration strategy and steady-state comparisons across fiscal regimes. Section 6 analyzes the dynamic effects of public investment shocks. Section 7 provides robustness checks, and Section 8 concludes.

2 Literature review

A large body of past literature examines the effects of fiscal policy on economic activity through output multipliers, with a focus on government purchases. Much of this work is limited to short- and long-run multiplier estimates without accounting for state dependency.

In contrast to dominating New Keynesian approaches, a smaller but influential strand of

the literature analyzes fiscal policy within flexible-price RBC frameworks. A foundational contribution is Baxter and King (1993), which serves as a key reference for this paper. The proposed model implements fiscal rules under the RBC framework, measuring macroeconomic impacts. Key ideas highlighted in this paper include intertemporal substitution and the household wealth effects, suggesting a crowding-out effect of government spending, as long-lived households anticipate higher future lump-sum taxes to sustain the tax base. Similar results were concluded by Edelberg et al. (1999), that longer working times boost output in the short run as a result of the negative wealth effect. Importantly, Baxter and King (1993) also shows that public investment can generate crowding-in effects by raising the productivity of public capital.

The effects of public investment also depend critically on how it is financed. While anticipated future taxation generates negative wealth effects, public investment may raise future returns by increasing productivity. Traum and Yang (2015) dive deeper into government debt financing and finds that the presence of higher returns to capital may outweigh the negative effects of future taxation, thereby inducing crowding-in of investment. This productivity-led crowding-in motivates an analysis of public investment multipliers across debt environments.

Motyovszki et al. (2024) suggest an increasing need for public investment in the EU to sustain or reduce current debt-to-GDP ratios. They argue that public investment can be self-financing by expanding the tax base through sustained GDP growth, generating positive long-run effects. However, short-run opposing effects were also seen as initial increases in primary deficits may outweigh the benefits of future expansion. This suggests that countries with higher debt ratios may have less fiscal room to implement such investments and realize their gains before debt constraints become binding. Related work from Leeper et al. (2010) uncovered the importance of timing through the US stimulus bills' implementation delay. They show that lags between initial public investment and the accumulation of productive public capital lead to small short-run multipliers.

Overall, the literature suggests that the effectiveness of public investment depends on debt-financing conditions, implementation dynamics, and existing debt levels. While public investment can crowd in private activity through capital accumulation, short-run multipliers are often limited when returns materialize slowly, particularly in high-debt environments facing fiscal constraints.

This paper contributes to the literature by examining debt-state dependent public investment multipliers. Instead of a homogeneous multiplier, the paper applies two distinctive high- and low-debt regimes to assess and contrast how debt conditions shape the effectiveness of public investment through a flexible-price RBC model.

3 Data

This paper uses annual data for European Union member states over the period 1995–2024. The dataset is constructed from publicly available sources, primarily Eurostat and the IMF, and covers key macroeconomic aggregates, summarized in Table 3, required for the calibration and evaluation of the model. The sample includes 21 EU countries, displayed in Figure 3, selected based on data availability and consistency over time. The sampled period faces a trade-off in selecting the data, due to gradual expansions of EU member states. Restricting the sample to countries with sufficiently long data series ensures comparability over time while maintaining broad regional representation.

The raw data consist of nominal series for gross domestic product, private consumption, private and public investment and capital, government debt, tax revenues, and wages, all of which are deflated using the implicit GDP deflator (2015 = 100) to obtain real variables.

For the purpose of model calibration, the country-level data are aggregated to an EU-wide economy by computing simple annual averages across countries for each variable. This aggregation strategy is consistent with the representative-agent framework of the model and allows us to abstract from cross-country heterogeneity while preserving the overall time-series properties of the European economy.

For comparative analysis, the dataset is split into high- and low-debt periods based on the debt-to-GDP ratio. As illustrated in Figure 4, the debt-to-GDP ratio begins to rise in 2008, likely explained by the impact of the Global Financial Crisis. Accordingly, 2008 is used as the cutoff year between the two regimes. Overall, the aggregated dataset provides a full overview of the relevant regions included in the calibration of parameters for quantitative analysis.

4 Model

This section develops a real business cycle framework in which productive public capital and government debt jointly interact with aggregate shocks to determine macroeconomic dynamics. The model is solved separately for two fiscal regimes: a low-debt period spanning 1995–2007 and a high-debt period covering 2008–2024 and therefore is designed to evaluate how the impact of public investment on output varies across these fiscal environments.

The model features a representative household, a representative firm and government. Prices and wages are flexible, markets are competitive, and uncertainty arises from exogenous technology and public investment shocks. The behavior of each economic agent, together with the resulting equilibrium conditions, is described in detail below.¹

4.1 Households

The economy is populated by a unit continuum of identical households that maximise expected lifetime utility represented by the Greenwood–Hercowitz–Huffman (GHH) preferences

$$\max_{\{C_t, N_t, K_{t+1}, I_t\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{\left(C_t - \psi \frac{N_t^{1+\phi}}{1+\phi} \right)^{1-\sigma}}{1-\sigma} \right],$$

where C_t denotes consumption, N_t labor supply, K_{t+1} next-period private capital, and I_t private investment. The parameter $\beta \in (0, 1)$ is the subjective discount factor, σ governs relative risk aversion, ϕ determines the Frisch elasticity of labor supply, and ψ scales the disutility of labor that ensures that hours worked are optimally chosen in the steady state. Each household faces a standard period budget constraint and capital accumulation equation following (Galí et al., n.d.) Galí et al. (2004);

$$C_t + I_t = w_t N_t + r_t K_t - T_t,$$

$$K_{t+1} = (1 - \delta) K_t + I_t,$$

where households supply labour N_t at wage w_t , own private capital K_t yielding rental income $r_t K_t$, and pay lump-sum taxes T_t . Consumption C_t , capital K_{t+1} and private investment I_t are subject to non-negativity constraints and labour supply satisfies $N_t \in [0, 1]$. Finally, the depreciation rate is constant and satisfies $0 < \delta < 1$.

¹All variables in the model are expressed in real terms.

The household solves its optimisation problem taking prices and taxes as given. Let λ_t denote the multiplier on the budget constraint and μ_t the multiplier on the capital accumulation constraint. The associated Lagrangian with the full derivation is outlined in the Appendix. The first-order conditions with respect to C_t , N_t , I_t , and K_{t+1} characterize optimal intertemporal consumption–savings behavior and intratemporal labor supply decisions. In particular, these conditions give rise to the Euler equation for private capital accumulation and the intratemporal labor supply condition:

$$\left(C_t - \psi \frac{N_t^{1+\phi}}{1+\phi}\right)^{-\sigma} = \beta \mathbb{E}_t \left[\left(C_{t+1} - \psi \frac{N_{t+1}^{1+\phi}}{1+\phi}\right)^{-\sigma} (1 + r_{t+1} - \delta) \right]$$

$$\psi N_t^\phi = w_t.$$

As implied by the first-order conditions, labor supply depends solely on the real wage and is independent of consumption. This motivates the use of GHH preferences, which eliminate wealth effects on labor supply and isolate the substitution channel, allowing a cleaner identification of the labor-market response to shocks.

4.2 Firms

A representative competitive firm produces a homogeneous good using private capital K_t , public capital K_t^G , and labour N_t . Output is given by the Cobb–Douglas production function

$$Y_t = A_t K_t^\alpha (K_t^G)^{\alpha^G} N_t^{1-\alpha-\alpha^G},$$

where $\alpha \in (0, 1)$ is the output elasticity of private capital and $\alpha^G \geq 0$ captures the productivity of public capital and $(\alpha^G + \alpha) \in (0, 1)$. Private capital and labor are hired in competitive factor markets at the real rental rate, r_t , and the real wage, w_t , respectively. Public capital is provided exogenously to the firm, can be used at no direct cost, and evolves according to the government’s investment decisions, as described in Section 3.3. The specification of the total factor productivity A_t , ensures that productivity fluctuations are persistent but mean-reverting, generating realistic business-cycle dynamics. Specifically, it evolves exogenously following a stationary AR(1) process

$$A_t = \exp(z_t),$$

$$z_t = \rho_z z_{t-1} + \varepsilon_t^z, \quad \varepsilon_t^z \sim \mathcal{N}(0, \sigma_z^2), \quad |\rho_z| < 1.$$

The objective of the representative firm is to maximize static profits by optimally choosing private capital K_t and labor N_t , while taking the rental rate of capital and the real wage as exogenous. Profit maximization implies that factor prices equal their respective marginal products. The firm's profit maximization problem and the associated first-order conditions are given by

$$\begin{aligned} \max \quad & \Pi_t = Y_t - w_t N_t - r_t K_t, \\ r_t = & \alpha \frac{Y_t}{K_t}, \quad w_t = (1 - \alpha - \alpha^G) \frac{Y_t}{N_t}, \end{aligned}$$

Under perfect competition, these conditions imply zero profits in equilibrium, and ensure that output is fully exhausted by factor payments.

4.3 Government

The government is modeled as a non-optimizing agent and does not solve an intertemporal welfare maximization problem. Instead, fiscal policy is conducted according to exogenous rules that govern public investment and its financing. For simplicity, the government neither consumes nor provides transfers. Its sole policy instrument is public investment I_t^G , which augments the stock of productive public capital. Public investment is financed through lump-sum taxes T_t and the issuance of one-period government bonds B_t .

Government bonds are assumed to be held by the external sector and therefore do not directly affect the optimization decisions of domestic households or firms. The government budget constraint is given by

$$I_t^G + (1 + i_t)B_t = T_t + B_{t+1},$$

where i_t denotes the real interest rate on government debt.

Public investment is subject to exogenous disturbances and follows an autoregressive process. Specifically, public investment evolves according to

$$\begin{aligned} I_t^G &= \bar{I}^G \exp(g_t), \\ g_t &= \rho_g g_{t-1} + \varepsilon_t^g, \quad \varepsilon_t^g \sim \mathcal{N}(0, \sigma_g^2), \quad |\rho_g| < 1, \end{aligned}$$

where $\bar{I}^G$ denotes the steady-state level of public investment.

Lastly, public investment accumulates into public capital according to the law of motion

$$K_{t+1}^G = (1 - \delta_G)K_t^G + I_t^G, \quad 0 < \delta_G < 1.$$

Fiscal rule

Fiscal policy is assumed to follow a simple tax-based feedback rule that stabilizes public debt around its target. In particular, the tax rate $\tau_t \equiv T_t/Y_t$ evolves according to

$$\tau_t = \bar{\tau} + \phi_b(b_t - \bar{b}) + \phi_y(\log Y_t - \log \bar{Y}),$$

where $b_t \equiv B_t/Y_t$ is the government debt-to-output ratio and $\bar{Y}$ denotes the steady-state level of output. The debt target $\bar{b}$ is treated as a regime-specific parameter, as the model is solved separately for low- and high-debt calibrations. The parameter $\phi_b > 0$ governs the strength of the fiscal response to deviations of debt from its target, thereby ensuring long-run debt sustainability, while $\phi_y \geq 0$ captures a countercyclical response of taxes to the output gap.

4.4 Market Clearing and Resource Constraint

Goods market clearing requires that aggregate output is fully absorbed by private consumption, private investment, and public investment, as well as by net government debt service. Accordingly, the aggregate resource constraint is given by

$$Y_t = C_t + I_t + I_t^G + (1 + i_t)B_t - B_{t+1},$$

where $(1 + i_t)B_t - B_{t+1}$ captures net payments associated with government debt.

Together with the government budget constraint, the laws of motion for private and public capital, and the fiscal rule governing taxation, the goods market clearing condition completes the set of equilibrium conditions and closes the model.

4.5 Steady state

To study how macroeconomic outcomes differ across low- and high-debt fiscal environments, the model is solved for two distinct steady states, each corresponding to a different calibration of the government debt target. Apart from this difference in fiscal parameters, the steady states are constructed using the same analytical procedure.

In the steady state, there are no stochastic disturbances to total factor productivity or public investment. All real variables are time-invariant, and expectations are constant. The steady-state equilibrium is therefore characterized by a system of nonlinear algebraic equations, which we summarize below.

Household optimality. The household's intertemporal Euler equation and intratemporal labor-leisure optimality condition reduce to

$$1 = \beta(1 + r - \delta), \quad (1)$$

$$\psi N^\phi = w. \quad (2)$$

Firm conditions. Profit maximization by competitive firms implies that factor prices equal their marginal products. In steady state, these conditions are given by

$$r = \alpha \frac{Y}{K}, \quad (3)$$

$$w = (1 - \alpha - \alpha^G) \frac{Y}{N} \quad (4)$$

Government. The steady-state stock of public capital and the government budget constraint satisfy

$$K^G = \frac{I^G}{\delta_G}, \quad (5)$$

$$I^G + iB = T, \quad (6)$$

where i denotes the steady-state interest rate on government debt. In steady state, the tax rate is constant and pinned down by the fiscal rule,

$$\tau = \bar{\tau}. \quad (7)$$

Resource constraint and capital accumulation. Aggregate output is allocated to private consumption, private investment, public investment, and interest payments on government debt:

$$Y = C + I + I^G + iB, \quad (8)$$

$$I = \delta K. \quad (9)$$

Given a set of structural parameters and steady-state calibration targets, the system of equilibrium conditions (1)–(9) uniquely determines the steady-state allocation for each fiscal regime. In particular, conditional on common preference and technology parameters and regime-specific targets for investment, capital, and government debt ratios, the model solves for the steady-state vector $(C, N, K, I, K^G, I^G, B, Y)$.

5 Calibration and Steady State

This section presents the calibration and steady-state computation underlying the analysis. The calibration strategy combines standard parameter values drawn from the macroeconomic literature with steady-state moments estimated from aggregated European Union data.

5.1 Calibration

The model is calibrated using a combination of structural parameters, steady-state normalizations, and regime-specific empirical targets. Structural parameters governing preferences and technology—namely the discount factor β , the real interest rate i , the private and public capital shares α and α^G , the coefficient of relative risk aversion σ , and the Frisch elasticity parameter ϕ —are held fixed across fiscal regimes and are taken from the macroeconomic literature. Preliminary data-based estimates were considered but yielded implausible values for some parameters, such as negative real interest rates or excessively large capital shares. We therefore adopt standard values to ensure economic plausibility and comparability.

Steady-state labor supply is normalized to $N = 1/3$, and total factor productivity is normalized to $A = 1$ (King & Rebelo, 1999). Given these normalizations and the intratemporal optimality condition implied by GHH preferences, the labor disutility parameter ψ is determined endogenously in steady state to ensure consistency between household labor supply and the real wage implied by firms' optimality conditions. As such, ψ acts as a scaling parameter and does not affect the model's dynamic responses.

The depreciation rates of private and public capital are disciplined by steady-state accumulation conditions. Specifically, private capital depreciation rate δ and public capital depreciation rate δ^G are inferred from steady-state investment–capital ratios. Differences in investment and capital ratios across fiscal regimes therefore translate into regime-specific

depreciation rates, while preserving consistency between capital stocks and investment flows.

Fiscal regimes differ through a set of regime-specific steady-state targets calibrated directly from the data. In particular, the steady-state ratios $(I/Y, I^G/Y, K/Y, K^G/Y, B/Y)$ are calibrated separately for the low- and high-debt regimes. These targets pin down the regime-specific debt target $\bar{b}$ and determine the $\bar{\tau}$ endogenously. All calibrated parameters and steady-state targets are summarized in Table 1.

Table 1: Structural parameters, normalizations, and steady-state calibration targets

Parameter	Description	Low Debt	High Debt
α	Private Capital Share	0.330	0.330
α^G	Public Capital Share	0.050	0.050
β	Time Preference	0.960	0.960
σ	Relative Risk Aversion	2.000	2.000
ϕ	Frisch Elasticity	1.000	1.000
i	Real Interest Rate	0.040	0.040
δ	Private Capital Depreciation	0.106	0.098
δ^G	Public Capital Depreciation	0.057	0.057
Normalization			
A	Total Factor Productivity	1.000	1.000
N	Labour Supply	0.333	0.333
Target			
I/Y	Private Investment to GDP	0.237	0.232
I^G/Y	Public Investment to GDP	0.034	0.033
K/Y	Private Capital to GDP	2.236	2.359
K^G/Y	Public Capital to GDP	0.593	0.577
B/Y	Debt to GDP	0.689	0.851
Model-implied objects			
ψ	Scale Parameter of Labor Disutility	2.736	2.810
$\bar{\tau}$	Tax rate	0.061	0.067

5.2 Steady-state comparisons

The steady-state allocations are determined by the calibrated parameter values together with the first-order conditions of the agent optimization problem. The model delivers regime-dependent steady states. Relative to the low-debt regime, the high-debt regime is characterized by higher levels of output, consumption, private capital, government debt, and taxes. These differences reflect the higher steady-state debt target in the high-debt regime and the associated fiscal adjustments required to sustain it.

Labor supply is identical across regimes by construction, reflecting the common normalization of steady-state hours. Public capital, as well as private and public investment, are quantitatively very similar across fiscal regimes, consistent with preference parameters, technology, and investment targets being held constant. Overall, the steady-state comparison confirms that the two regimes differ primarily in their fiscal positions, while preferences, technology, labor supply, and investment behavior remain unchanged. This aligns with prior economic intuition and provides a coherent and internally consistent baseline for analyzing how public investment shocks differ across fiscal environments. Table 4 in the Appendix reports the steady-state values of the endogenous variables for both fiscal regimes.

6 Dynamic Effects of Public Investment

This section addresses the core research question of the paper by examining the effects of public investment shocks on output through the fiscal multiplier. We assess these effects in two complementary ways. First, we analyze impulse response functions to provide a visual and dynamic characterization of the economy’s response to a public investment shock. Second, we quantify the impact through numerical estimates of various fiscal multipliers, allowing for a systematic comparison across fiscal regimes.

6.1 Impulse Response Functions

To assess the dynamic effects of fiscal policy and to analyze how public investment shocks propagate through the economy under different fiscal conditions, we compute impulse response functions (IRFs) to a positive public investment shock. The shock is specified as an exogenous one-percent transitory increase in public investment, with an identical initial magnitude across fiscal regimes. The persistence of the public investment process, ρ_g , is estimated by Ordinary Least Squares and is set to 0.967 in the low-debt regime and 0.919 in the high-debt regime. This specification allows for a clean comparison of dynamic responses across low- and high-debt environments.

It is important to note that, while the shock itself is normalized to be comparable across regimes, the fiscal policy environment differs substantially. Government policy follows a fiscal rule whose parameters vary by regime, with the debt and output feedback coefficients, ϕ_b and

ϕ_y , as key drivers of the heterogeneous responses. The estimated values of ϕ_b are 0.109 and 0.104 in the low- and high-debt regimes, respectively, while the corresponding values of ϕ_y are 0.144 and 0.064. These coefficients are obtained from simple linear regressions estimated separately over the two subsamples.

Figure 1 reports the IRFs of public investment, output, consumption, private investment, public capital, and labor for both fiscal regimes. In both cases, the public investment shock generates a persistent increase in public capital, reflecting the gradual accumulation of productive public infrastructure. However, the responses of aggregate macroeconomic variables differ markedly across regimes.

In the low-debt regime, the increase in public investment leads to a strong and persistent expansion in output. The higher stock of public capital raises the marginal productivity of private inputs, stimulating labor demand and increasing real wages. As a result, labor supply rises and amplifies the output response. Private investment initially declines due to short-run resource reallocation toward public investment, but subsequently recovers and increases as higher productivity and improved factor returns encourage capital accumulation. Consumption exhibits a modest decline on impact, followed by a sustained expansion as income and productivity effects outweigh the initial crowding-out forces.

In contrast, the macroeconomic response in the high-debt regime is substantially attenuated. Although public capital still increases following the shock, the resulting output response is weaker and less persistent. Elevated government debt necessitates stronger fiscal adjustment through taxes, which dampens disposable income and weakens incentives for private investment and labor supply. Consequently, the expansion in private investment and employment is muted, limiting the amplification mechanisms observed in the low-debt regime. Consumption increases by a smaller magnitude and converges back to its steady state more rapidly.

6.2 Fiscal Multipliers

To quantify the effectiveness of public investment under different fiscal conditions, we compute output multipliers that summarize the impulse response analysis. These multipliers capture short- and medium-run effects and provide a compact metric for comparing fiscal transmission across debt regimes.

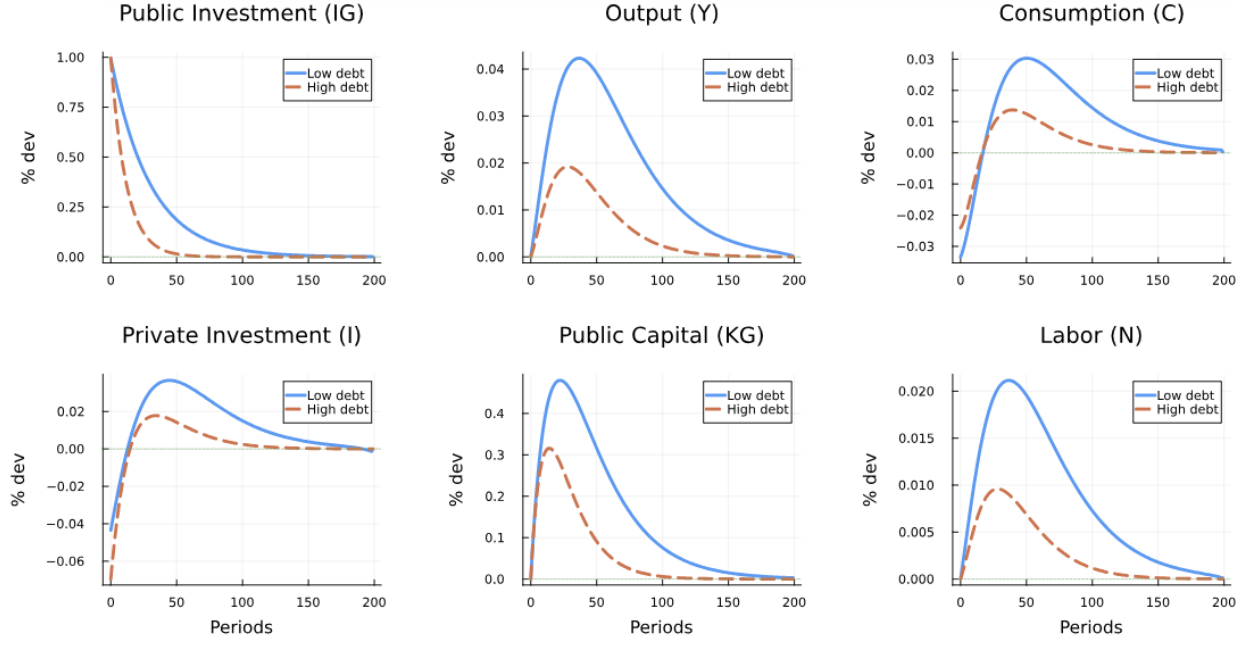


Figure 1: Impulse Response Functions to Public Investment Shocks

Table 2 reports four commonly used multiplier measures. The impact multiplier measures the contemporaneous change in output relative to the change in public investment at the time of the shock. The one-year-ahead multiplier captures the output response one period after the shock. The cumulative multiplier is defined as the ratio of the cumulative change in output to the cumulative change in public investment over the simulation horizon. Finally, the peak multiplier records the maximum ratio of output to public investment responses attained along the impulse response path.

The results reveal pronounced differences in the effectiveness of public investment across fiscal regimes. In both regimes, the impact multiplier is zero, reflecting the gradual nature of capital accumulation. Differences emerge rapidly over time. In the low-debt regime, the one-year-ahead multiplier reaches 0.057, more than twice the corresponding value in the high-debt regime. These disparities become more pronounced at longer horizons. The cumulative multiplier amounts to 2.945 in the low-debt regime, compared to 2.622 in the high-debt regime. Moreover, the peak multiplier exceeds unity in the low-debt regime, indicating a strong amplification of public investment, while it remains well below one in the high-debt regime, reflecting the dampening role of fiscal adjustment in high-debt environments.

These findings further support our intuition and indicate that public investment is considerably more effective at stimulating economic activity when public debt is low. Greater fiscal

Table 2: Fiscal Multipliers

Metric	Low Debt	High Debt
Impact multiplier	0.000	0.000
1-year-ahead multiplier	0.057	0.027
Cumulative multiplier	2.945	2.622
Peak multiplier	1.255	0.581

space allows the productivity-enhancing effects of public capital to translate into stronger private-sector responses, particularly through higher labor supply and private investment. In contrast, high public debt dampens the transmission of public investment by requiring stronger fiscal adjustments, which attenuate the expansionary effects on output. Overall, the multiplier estimates reinforce the impulse response analysis and highlight the importance of fiscal conditions in shaping the macroeconomic returns to public investment.

7 Robustness check

As a robustness check, we assess the model from two complementary perspectives. First, we analyze the dynamic responses to a standard total factor productivity (TFP) shock to verify that the model generates conventional business-cycle dynamics. Second, we apply the Simulated Method of Moments (SMM) to evaluate the model’s empirical performance and to identify its strengths and potential shortcomings.

7.1 Dynamic Effects of Total Factor Productivity

Figure 2 reports impulse response functions to a positive one-percent total factor productivity (TFP) shock for low-debt and high-debt regimes. The shock follows an autoregressive process estimated using the Solow residual methodology and is identical across regimes, with persistence $\rho_z = 0.94$. In contrast to the public investment shock, the TFP disturbance directly affects production and therefore generates an immediate and pronounced increase in output, consumption, private investment, labor, and private capital.

The dynamic responses are highly similar across fiscal regimes. Output rises sharply on impact and gradually converges back to its steady state as productivity mean-reverts. Consumption and private investment increase strongly, reflecting higher income and improved

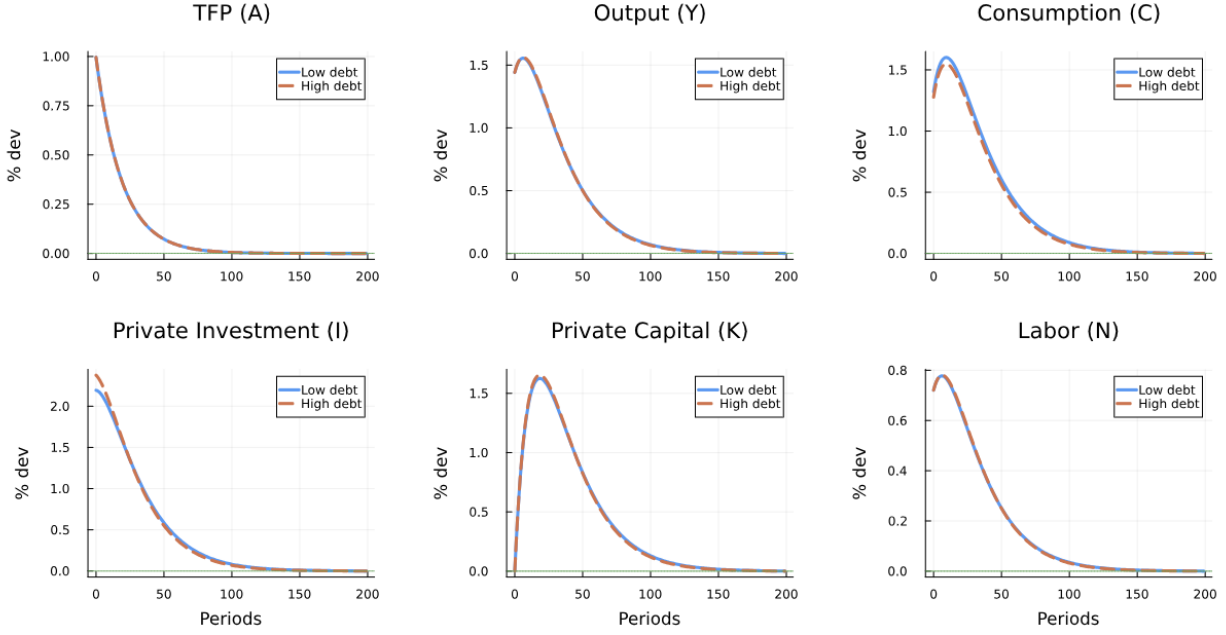


Figure 2: Impulse Response Functions to Total Factor Productivity

intertemporal incentives. Labor supply responds positively due to higher real wages, while private capital accumulates gradually as firms expand investment in response to higher productivity. These dynamics are consistent with standard real business cycle mechanisms and confirm that the model behaves as expected in response to supply-side shocks.

Importantly, differences between the low-debt and high-debt regimes are negligible across all variables. This near-equivalence indicates that government debt plays only a limited role in shaping the transmission of TFP shocks, as such shocks do not directly affect fiscal balances or trigger immediate fiscal adjustment. Consequently, the strong regime dependence observed after public investment shocks does not arise in response to productivity disturbances. This supports the interpretation that fiscal space, rather than underlying productivity dynamics, is the primary driver of heterogeneous public investment effects across debt regimes.

7.2 Simulated Method of Moments

This section evaluates the empirical performance of the model using the Simulated Method of Moments. The objective of this exercise is not to exactly replicate all features of the data, but rather to assess whether the model is able to reproduce key second-moment properties of macroeconomic aggregates that are central to business-cycle dynamics. In particular, we focus on the relative volatilities of consumption, private investment, and public investment

with respect to output, as well as their contemporaneous correlations with output. These moments are computed separately for low- and high-debt regimes in order to assess whether the model captures differences in macroeconomic dynamics across fiscal environments.

The model is simulated using stochastic realizations of total factor productivity and public investment shocks, whose persistence and variance are estimated via SMM. Simulated series are generated from the linearized model and filtered in a manner consistent with the empirical data. Table 5 in the Appendix reports the corresponding empirical and model-implied moments.

Overall, the model performs well in reproducing the cyclical co-movement between output, consumption, and private investment across both debt regimes. In the data, consumption is strongly procyclical, with correlations of approximately 0.95–0.97 with output. The model closely matches these correlations in both low- and high-debt environments, indicating that the consumption-smoothing and intertemporal substitution mechanisms embedded in the model are effective in propagating aggregate shocks.

In terms of relative volatility, the model underpredicts the variability of consumption and private investment with respect to output. While consumption volatility in the data ranges between 0.96 and 1.12 across regimes, the model generates substantially lower values, around 0.44–0.48. Similarly, private investment volatility is notably higher in the data than in the model, particularly in the low-debt regime. These discrepancies are consistent with limitations of standard DSGE models, which tend to smooth consumption excessively and dampen investment responses unless additional frictions or adjustment costs are introduced.

The largest quantitative divergence between the model and the data arises for public investment. Empirically, public investment is moderately procyclical and exhibits volatility comparable to that of output. By contrast, the model generates extremely low volatility for public investment and near-zero correlation with output in both debt regimes. This result reflects the relatively small estimated variance of public investment shocks and the fact that government investment dynamics in the model are largely policy-driven rather than endogenous responses to aggregate fluctuations. Importantly, this finding should be interpreted as a structural implication of the estimated shock process rather than as a failure of the transmission mechanism itself.

8 Conclusion

This paper studies how public investment affects the multiplier under high- and low-debt periods through a flexible price RBC framework in the EU countries. Impulse response analysis shows that TFP shocks generate relatively similar dynamics across debt regimes, whereas public investment shocks lead to stronger and more persistent responses in low-debt states, with larger deviations from the steady state. Consistent with these responses, larger multipliers are present, confirming that low-debt states exhibited higher impact across all three time horizons, as well as a higher peak multiplier. Simulated moments further indicate persistent investment effects in both regimes, alongside greater volatility in high-debt economies, reflecting that the model matches real-world behaviors. Together, these findings highlight the central role of initial debt conditions in shaping the effectiveness of public investment policies.

While the model provides a strong foundation through the inclusion of fiscal rules in public investment, several extensions remain relevant. Allowing for distortionary taxation, such as labor taxes that reduce marginal returns, would help assess efficiency losses and their implications for fiscal effectiveness. In addition, incorporating monetary policy would allow further analysis of interactions between debt and interest rates. This may help shed light on whether monetary responses can mitigate weaker fiscal effects in high-debt states.

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9 Appendix

A Household Lagrangian Optimization - First-Order Conditions

Let X_t denote

$$X_t \equiv C_t - \psi \frac{N_t^{1+\phi}}{1+\phi}. \quad (10)$$

Then the Lagrangian becomes

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{X_t^{1-\sigma}}{1-\sigma} + \lambda_t (w_t N_t + r_t K_t - T_t - C_t - I_t) + \mu_t ((1-\delta)K_t + I_t - K_{t+1}) \right]. \quad (11)$$

From $u(X_t) = \frac{X_t^{1-\sigma}}{1-\sigma}$ we have

$$u_X(X_t) = X_t^{-\sigma}. \quad (12)$$

Moreover,

$$\frac{\partial X_t}{\partial C_t} = 1, \quad \frac{\partial X_t}{\partial N_t} = -\psi N_t^\phi. \quad (13)$$

First-order conditions

(i) **Consumption** C_t . Differentiate (11) with respect to C_t :

$$0 = \frac{\partial}{\partial C_t} \left[\beta^t \left(\frac{X_t^{1-\sigma}}{1-\sigma} + \lambda_t (\dots - C_t - \dots) \right) \right] \quad (14)$$

$$= \beta^t \left(u_X(X_t) \frac{\partial X_t}{\partial C_t} - \lambda_t \right) = \beta^t (X_t^{-\sigma} - \lambda_t). \quad (15)$$

Hence,

$$\lambda_t = X_t^{-\sigma} = \left(C_t - \psi \frac{N_t^{1+\phi}}{1+\phi} \right)^{-\sigma}. \quad (16)$$

(ii) **Labor** N_t . Differentiate with respect to N_t :

$$0 = \beta^t \left(u_X(X_t) \frac{\partial X_t}{\partial N_t} + \lambda_t w_t \right) = \beta^t \left(X_t^{-\sigma} (-\psi N_t^\phi) + \lambda_t w_t \right). \quad (17)$$

Thus,

$$\boxed{\lambda_t w_t = \psi N_t^\phi X_t^{-\sigma}}. \quad (18)$$

Using (16) (i.e., $\lambda_t = X_t^{-\sigma}$), this simplifies to the standard GHH intratemporal condition:

$$w_t = \psi N_t^\phi. \quad (19)$$

(iii) Investment I_t . Differentiate with respect to I_t :

$$0 = \beta^t (-\lambda_t + \mu_t), \quad (20)$$

implying

$$\mu_t = \lambda_t. \quad (21)$$

(iv) Next-period capital K_{t+1} . K_{t+1} appears in period t through $-\mu_t K_{t+1}$ and in period $t+1$ through $r_{t+1} K_{t+1}$ in the budget constraint and $(1-\delta)K_{t+1}$ in the capital law of motion.

Therefore,

$$0 = \frac{\partial \mathcal{L}}{\partial K_{t+1}} = \mathbb{E}_0 \left[\beta^t (-\mu_t) + \beta^{t+1} (\lambda_{t+1} r_{t+1} + \mu_{t+1} (1-\delta)) \right]. \quad (22)$$

Divide by β^t :

$$\mu_t = \beta \mathbb{E}_t [\lambda_{t+1} r_{t+1} + \mu_{t+1} (1-\delta)]. \quad (23)$$

Using (21) (i.e., $\mu_t = \lambda_t$ and $\mu_{t+1} = \lambda_{t+1}$), we obtain the Euler equation:

$$\lambda_t = \beta \mathbb{E}_t [\lambda_{t+1} (r_{t+1} + 1 - \delta)]. \quad (24)$$

Substituting $\lambda_t = X_t^{-\sigma}$ from (16) yields the Euler equation in quantities:

$$\left(C_t - \psi \frac{N_t^{1+\phi}}{1+\phi} \right)^{-\sigma} = \beta \mathbb{E}_t \left[\left(C_{t+1} - \psi \frac{N_{t+1}^{1+\phi}}{1+\phi} \right)^{-\sigma} (r_{t+1} + 1 - \delta) \right]. \quad (25)$$

B Firm Optimization - First-Order Conditions

$$\max \quad \Pi_t = Y_t - w_t N_t - r_t K_t \quad (26)$$

$$\text{subject to} \quad Y_t = A_t K_t^\alpha (K_t^G)^{\alpha^G} N_t^{1-\alpha-\alpha^G}, \quad (27)$$

Using (27), the marginal products of private capital and labor are

$$\frac{\partial Y_t}{\partial K_t} = A_t (K_t^G)^{\alpha^G} N_t^{1-\alpha-\alpha^G} \cdot \alpha K_t^{\alpha-1} = \alpha \frac{Y_t}{K_t}, \quad (28)$$

$$\frac{\partial Y_t}{\partial N_t} = A_t K_t^\alpha (K_t^G)^{\alpha^G} \cdot (1 - \alpha - \alpha^G) N_t^{-\alpha-\alpha^G} = (1 - \alpha - \alpha^G) \frac{Y_t}{N_t}. \quad (29)$$

The first-order conditions are obtained by differentiating (26) with respect to K_t and N_t :

$$\frac{\partial \Pi_t}{\partial K_t} = \frac{\partial Y_t}{\partial K_t} - r_t = 0 \quad \implies \quad r_t = \frac{\partial Y_t}{\partial K_t} = \alpha \frac{Y_t}{K_t}, \quad (30)$$

$$\frac{\partial \Pi_t}{\partial N_t} = \frac{\partial Y_t}{\partial N_t} - w_t = 0 \quad \implies \quad w_t = \frac{\partial Y_t}{\partial N_t} = (1 - \alpha - \alpha^G) \frac{Y_t}{N_t}. \quad (31)$$

Thus, under perfect competition, factor prices equal marginal products:

$$r_t = \alpha \frac{Y_t}{K_t}, \quad w_t = (1 - \alpha - \alpha^G) \frac{Y_t}{N_t}.$$

C Tables and Figures

Table 3: Description of variables

Variable	Description
GDP	Gross domestic product
Consumption	Aggregate private consumption
Public_Investment	Government investment expenditure
Private_Investment	Private fixed capital formation
Debt	Government gross debt
Taxes	Total government tax revenues
Wages	Aggregate real wage bill
Private_Capital	Private capital stock
Public_Capital	Public capital stock
Labour	Total hours worked
DebttoGDP	Government debt-to-GDP ratio
Interest	Effective interest rate on government debt



Figure 3: European countries included in the dataset

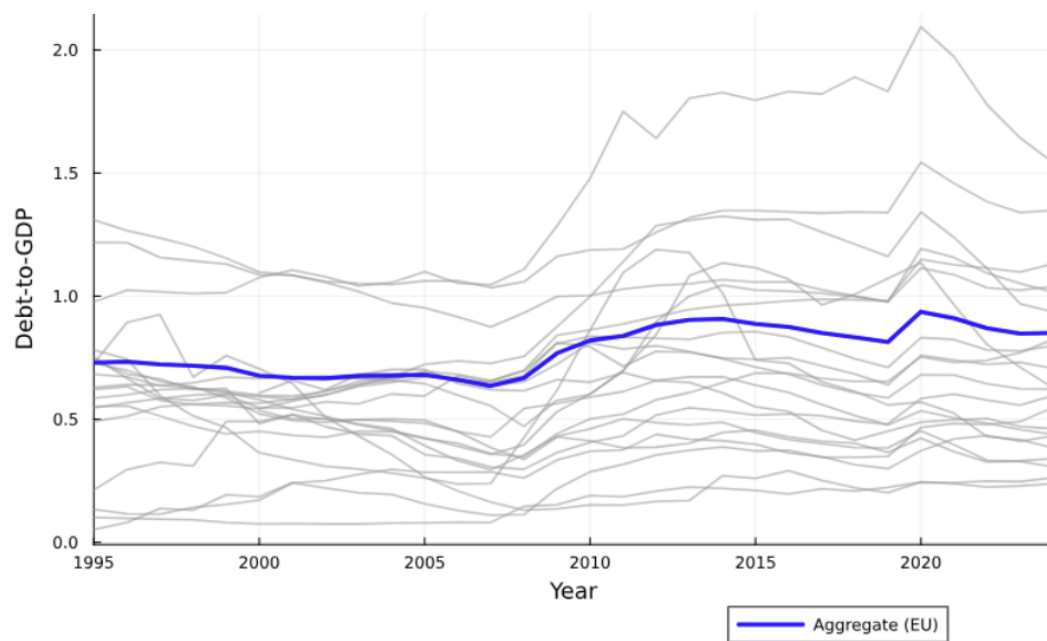


Figure 4: Aggregated Debt-to-GDP ratio

Table 4: Steady-state endogenous variables

Variable	Description	Low Debt	High Debt
Y	Output (GDP)	0.490	0.504
C	Consumption	0.344	0.353
I	Private investment	0.116	0.117
I^G	Public investment	0.016	0.017
N	Labour	0.333	0.333
K	Private capital	1.096	1.188
K^G	Public capital	0.290	0.290
B	Debt	0.338	0.428
T	Taxes	0.030	0.034

Table 5: Matching moments: Volatilities and Correlations Relative to Output

Variable	Volatility				Correlation			
	Low Debt		High Debt		Low Debt		High Debt	
	Data	Model	Data	Model	Data	Model	Data	Model
Consumption (C)	0.96	0.48	1.12	0.44	0.97	0.98	0.95	0.99
Private Investment (I)	1.90	1.04	1.63	1.13	0.81	0.97	0.88	0.98
Public Investment (I^G)	1.43	0.05	1.39	0.04	0.53	0.02	0.48	0.04